claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-a62dac359b7a42c94.jsonl`
- SHA-256: `39fd1f887436ce2a16423869689a961e838b40996e3e12acfe29a65d6e5c3339`
- Source modified: `2026-06-16T03:20:06+00:00`
- Imported at: `2026-07-05T16:48:27+00:00`
- project: `wf_3d536535-241`
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`

Transcript

1. user (2026-06-16T03:19:35.882Z)

Source Extractor

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Research question: "Research question: **What concrete, recent (2022?2025) techniques and
evaluation practices should a local, commercially-licensed badminton rally-detector adopt to
close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16
golden videos)?** Produce an adversarially-verified, cited report organized around the four
dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over
generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):
- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a
heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff"
confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits
far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it
does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active"
iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold;
active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary
split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves +
knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-
frame visibility).
- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom
licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM
(Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference;
minors excluded from any training pool; runs on a single local GPU.
- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture
skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) +
late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet,
See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation,
tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-
one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-
count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE,
a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment
leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal
windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-
window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ?
what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented
actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression
and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression),
BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads),
smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k,
```

edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RLOOCV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible."

Fetch and extract key claims from this source:
URL: https://news.ycombinator.com/item?id=35375670
Title: OpenAI Terms of Service forbid training competitor models via their ML outputs (HN discussion)
Found via: Knowledge-distillation licensing boundary + label-noise from teacher search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
   - be a concrete, checkable statement (not vague generalities)
   - include a direct quote from the source as support
   - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-16T03:19:38.635Z)

I'll fetch the page content first.

3. user (2026-06-16T03:19:50.336Z)

OpenAI ToS Claims Analysis

Initial Claim

The thread opens with a user stating: "OpenAI Terms of service forbid training competitor models via their ML outputs." However, **no actual OpenAI ToS text is quoted in this entire Hacker News discussion**.

What Commenters Say

On Enforceability:
- One commenter notes that "Google has no contract with OpenAI" and questions enforceability since they scraped from third-party sites
- Another points out: "The recent HiQ vs LinkedIn case would seem to make this ToS unenforceable, unless Google actually created a user account on ShareGPT"

On Common Practice:
- A commenter states: "OpenAI and Google both scour the web for content, period" and argues neither company respects all copyright/ToS equally
- Another challenges the distinction: "So there really isn't that much difference between the two"

On Hypocrisy:
- Multiple users note OpenAI trained on copyrighted works without permission, making restrictions on their outputs seem contradictory
- One observes: "I love it how they don't want others to use their model output but they have no qualms about training their model on the copyrighted works of others"

Key Gap
The discussion contains **no actual quoted language** from OpenAI's Terms of Service, making it impossible to verify the claim against source documentation.

Publication Date: March 30-31, 2023

4. assistant (2026-06-16T03:19:54.633Z)

This is a forum discussion (Hacker News). The fetch succeeded. Let me assess: it's a forum thread from March 2023 discussing OpenAI's ToS restriction on using outputs to train competing models ? relevant to the research question's governance/distillation-boundary dimension (Dimension 3), but the source contains only commenter opinions, not authoritative ToS text. I'll extract the falsifiable claims it does support.

5. user (2026-06-16T03:19:59.297Z)

Structured output provided successfully

6. assistant (2026-06-16T03:20:06.322Z)

Source processed. It is a Hacker News forum thread (March 30, 2023) discussing OpenAI's ToS restriction on training competitor models from its outputs.

Key findings relevant to the research question's Dimension 3 (legally-clean teacher exploitation / distillation boundaries):

- The thread confirms the existence of the policy premise (OpenAI ToS forbids using its outputs to train competing models), which directly supports the project's hard constraint of never training owned weights on paid-LLM outputs while permitting inference-time use.
- However, the source quality is low (forum opinion). Critically, the discussion contains NO actual quoted ToS text ? commenters debate enforceability (citing hiQ v. LinkedIn) and OpenAI's asymmetric copyright stance, but provide no verifiable primary-source clause.

Caveats flagged: All three extracted claims are commenter assertions, not authoritative legal/ToS text. The central governance claim should be verified against the actual OpenAI Terms of Use (a primary source), not this thread. The enforceability and copyright-hypocrisy claims are tangential opinion and should not be relied on for the report's governance section.